

A Multi-View Photometric Stereo Pipeline for Specular 3D Fruit Reconstruction

Ariel Zuniga-Santana¹ - Rodrigo Verschae¹ - Gabriele Facciolo² - Shohei Nobuhara³

¹Institute of Engineering Sciences, Universidad de O'Higgins, Chile

²Universite Paris-Saclay, CNRS, ENS Paris-Saclay, Centre Borelli, France

³Institut Universitaire de France (IUF)

⁴Information and Human Sciences, Kyoto Institute of Technology, Japan

Motivation

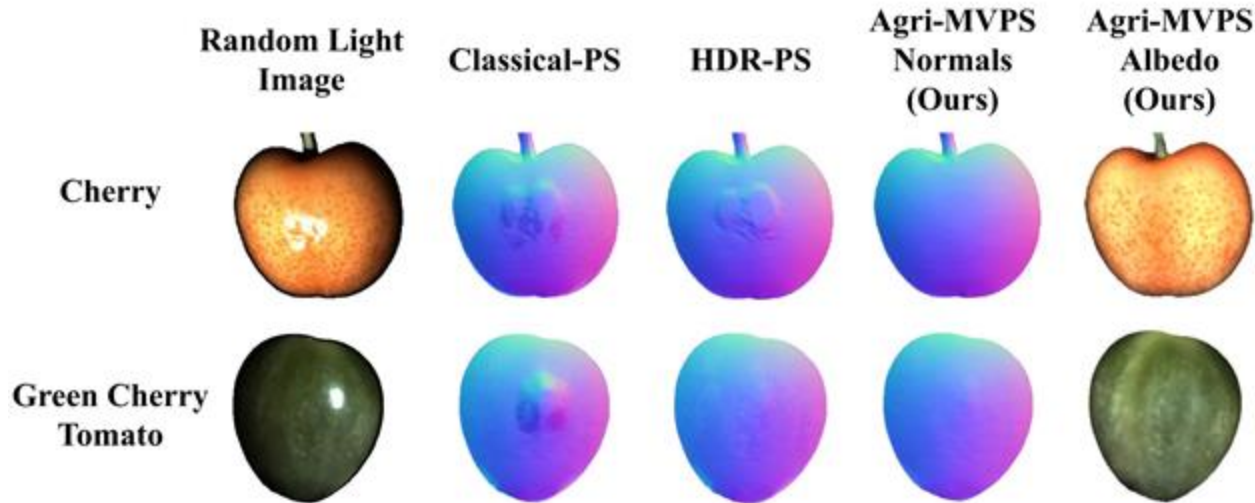
Small Glossy Fruits Challenge Classical 3D Reconstruction

Small glossy fruits exhibit smooth surfaces, low texture, and strong specular reflections. Classical Multi-View Stereo often produces incomplete and unstable geometry.



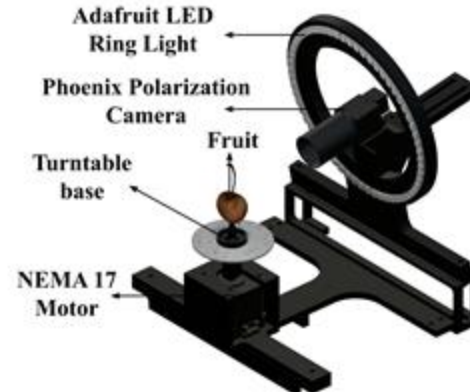
Classical PS Assumptions Break Under Close Lighting

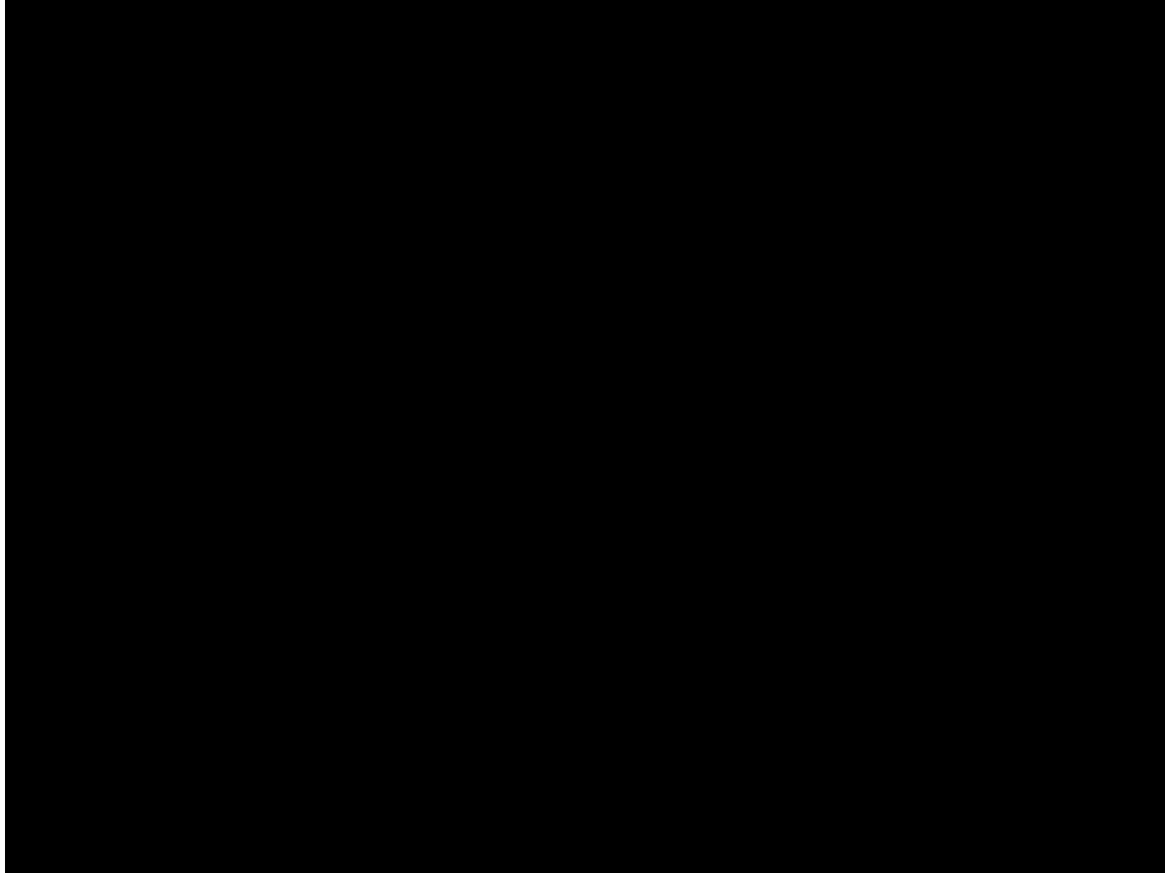
With near-field illumination, glossy fruits exhibit strong, spatially varying highlights and saturation, making Lambertian-based normal estimation unstable.



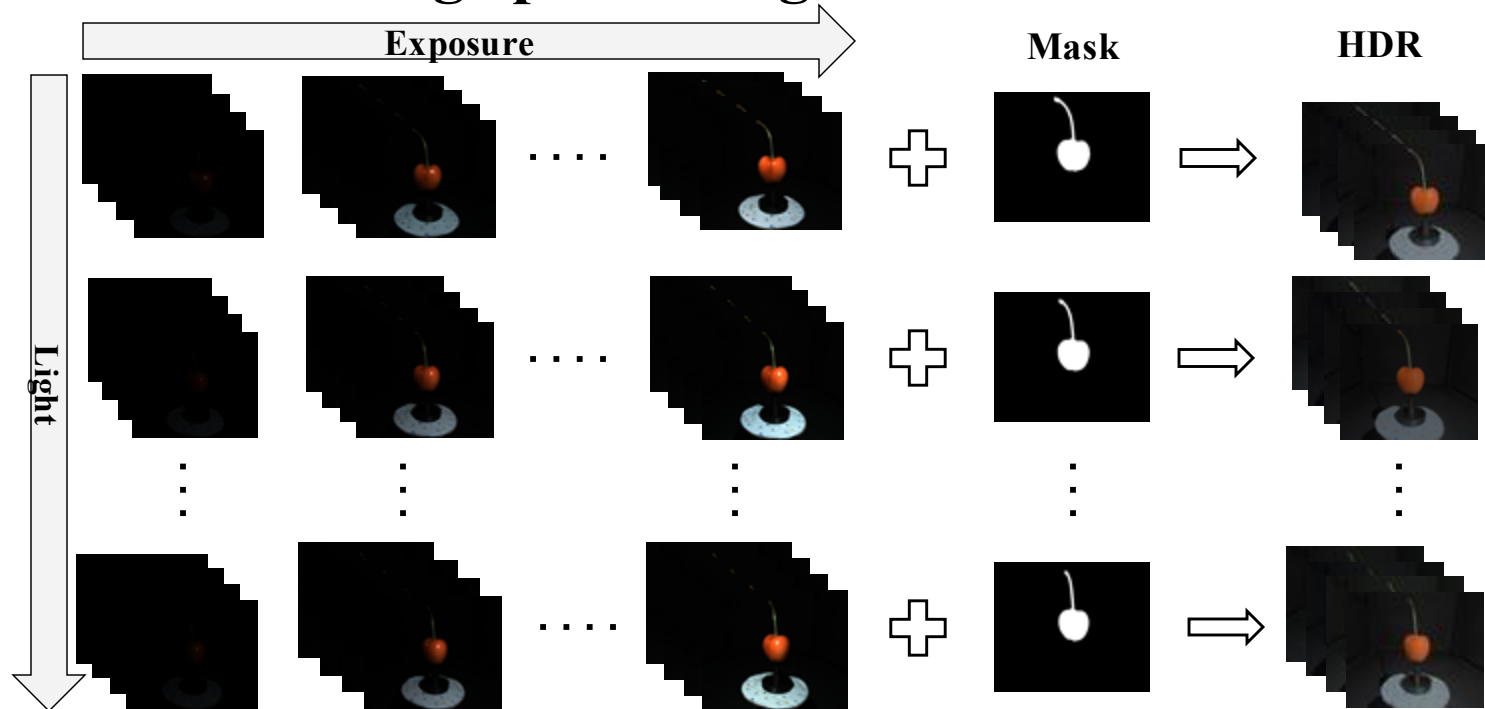
Data Acquisition

For each fruit, the turntable rotates in 18° increments with 20 viewpoints. At each viewpoint, polarized images are captured sequentially for 26 LEDs with a Phoenix Polarization Camera (BayerRG). This procedure is repeated for six exposure times (4.8, 2.4, 1.2, 0.6, 0.3, and 0.15 s).



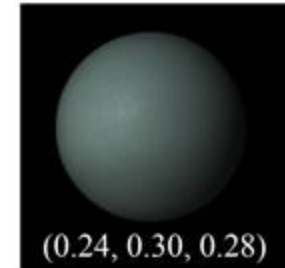
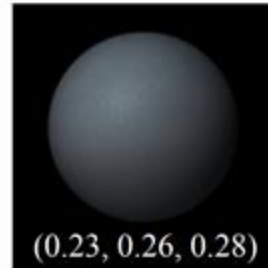
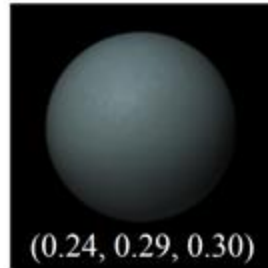
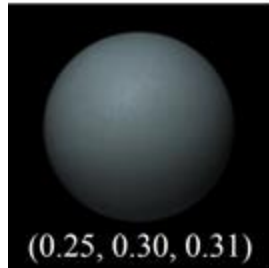


Polarized HDR image processing

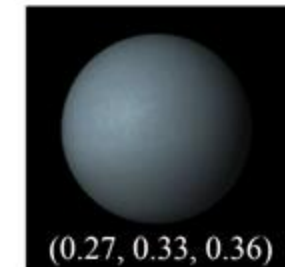
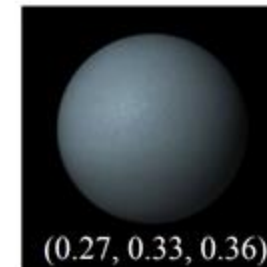
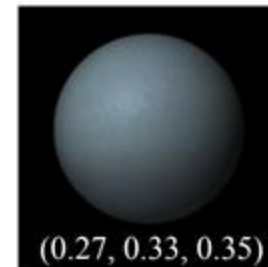
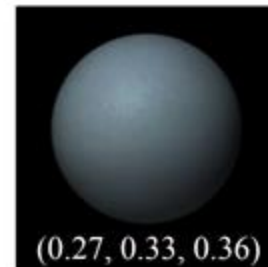
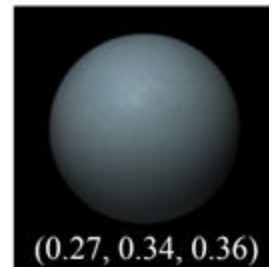


Light Intensities Calibration

Diffuse Ball Before Calibration

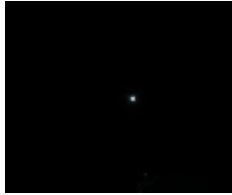
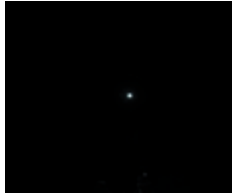
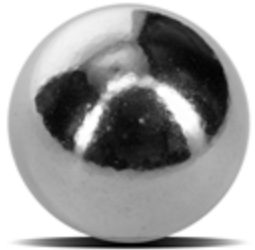


Diffuse Ball After Calibration

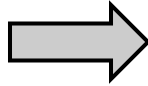
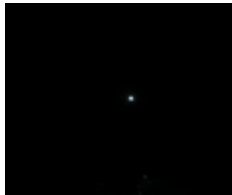


Light Directions Calibration

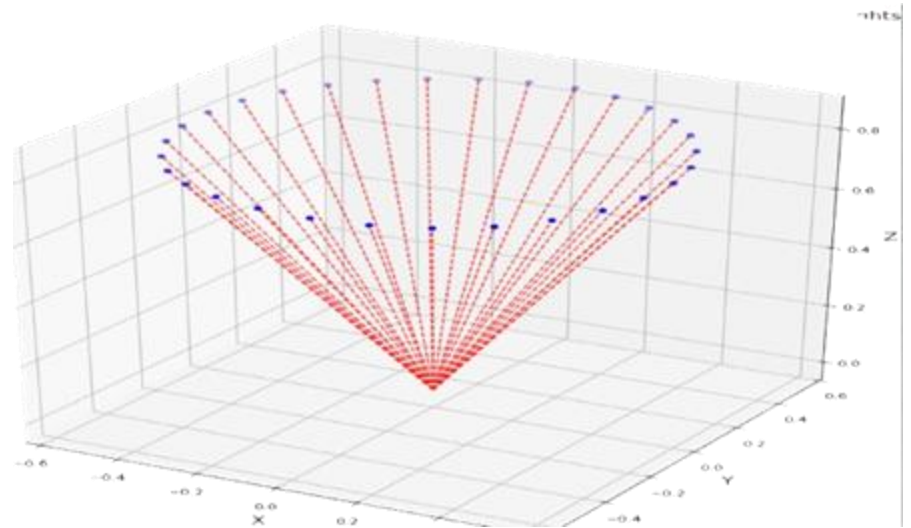
Metal Ball



⋮



Estimated Light Directions



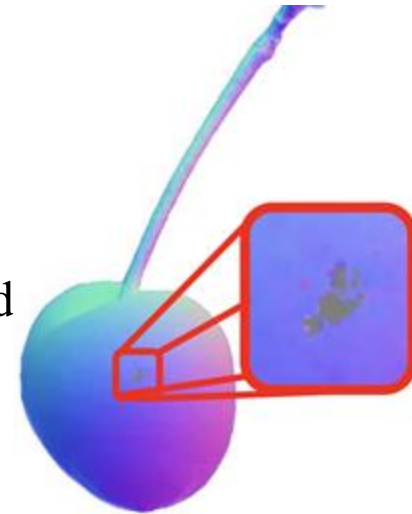
Normals Estimation

We estimate surface normals assuming unit albedo ($\rho = 1$)

To reduce highlight influence:

- Only the red channel is used
- For each pixel, intensities above the 90th percentile are discarded
- The number of valid lights varies per pixel

$$\mathbf{n}_i = \frac{(\mathbf{L}_i^\top \mathbf{L}_i)^{-1} \mathbf{L}_i^\top \mathbf{I}_i^{\text{scaled}}}{\|(\mathbf{L}_i^\top \mathbf{L}_i)^{-1} \mathbf{L}_i^\top \mathbf{I}_i^{\text{scaled}}\|}$$



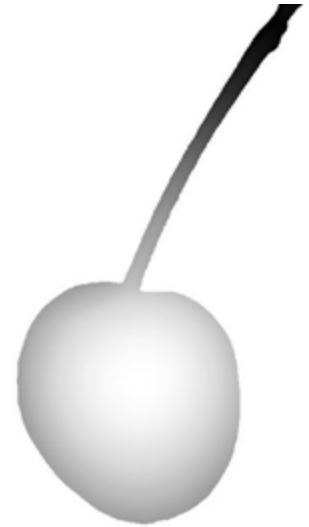
Refining Normals

To reduce noise in normals, depth is computed as the gradients.

$$p(i, j) = -\frac{n_x(i, j)}{n_z(i, j)}, \quad q(i, j) = -\frac{n_y(i, j)}{n_z(i, j)}$$

Then, two restrictions are imposed and a linear system is solved with least squares.

$$z(i, j+1) - z(i, j) = p(i, j), \quad z(i+1, j) - z(i, j) = q(i, j)$$

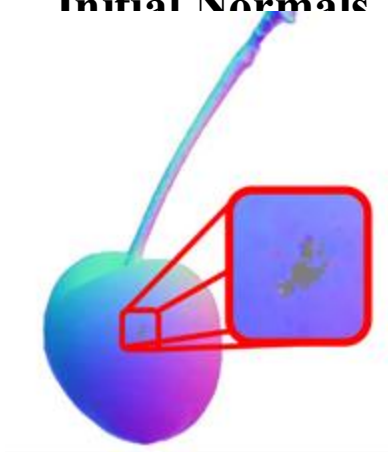


Refining Normals

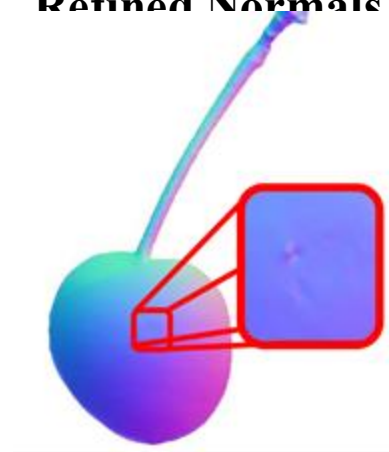
Normals are obtained from depth.

$$\mathbf{n}_{i,\text{refined}} = \frac{(-\partial z_i / \partial x, -\partial z_i / \partial y, 1)}{\|(-\partial z_i / \partial x, -\partial z_i / \partial y, 1)\|}$$

Initial Normals



Refined Normals



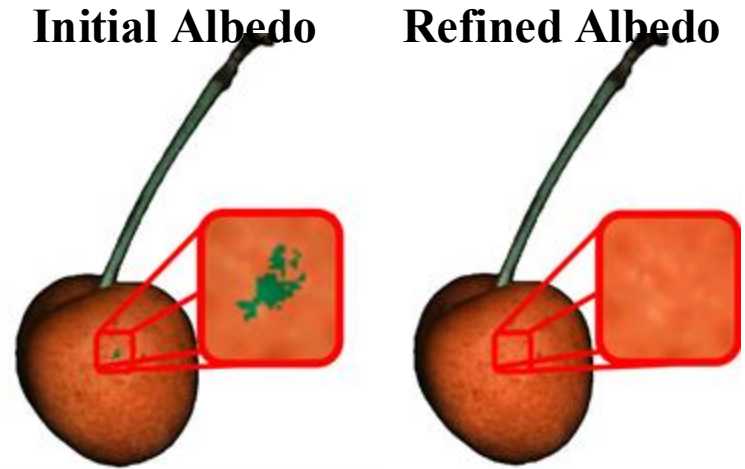
Albedo Estimation

For each pixel using only valid lights and applying a robust weighting with

$$w_{i,d,c} = \exp \left(- \frac{\max(I_{i,d,c}^{\text{scaled}} - I_{\text{med},i,c}, 0)^2}{2\sigma_{i,c}^2} \right)^\beta$$

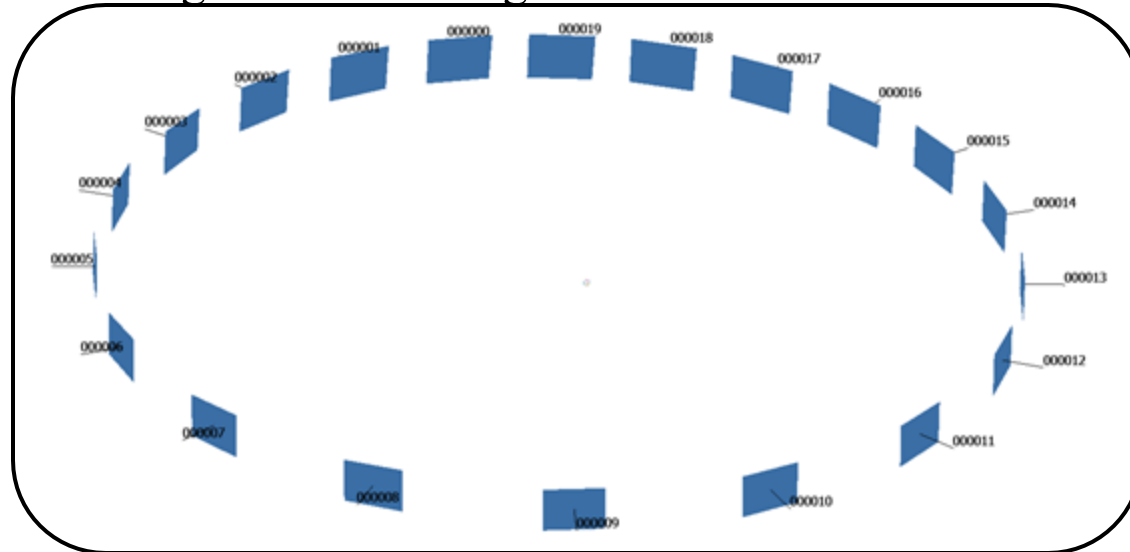
Albedo is estimated with a weighted least-squares formulation:

$$\rho_{i,c} = \frac{\sum_{d \in V_i} w_{i,d,c} I_{i,d,c}^{\text{scaled}} (\mathbf{n}_{i,\text{refined}} \cdot \mathbf{l}_d)}{\sum_{d \in V_i} w_{i,d,c} (\mathbf{n}_{i,\text{refined}} \cdot \mathbf{l}_d)^2}$$



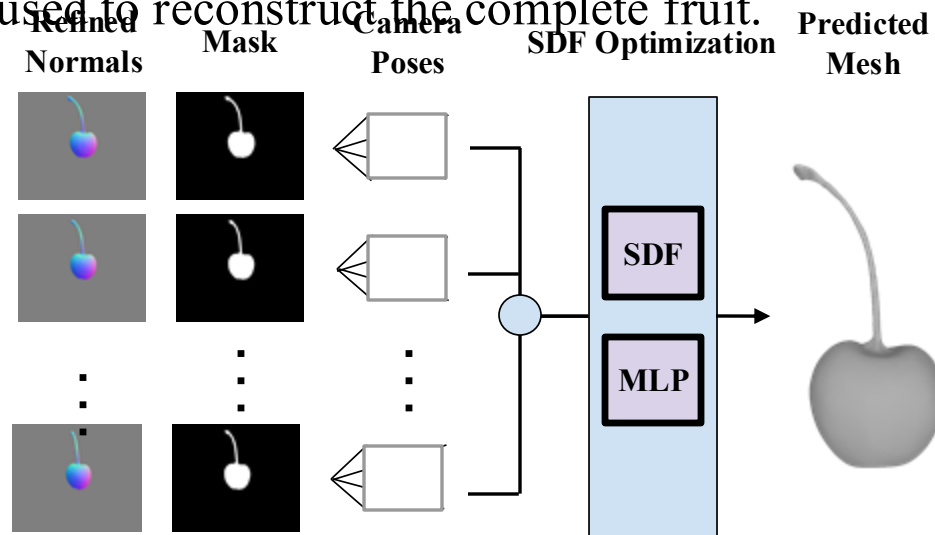
3D fruit reconstruction

Cameras extrinsics and intrinsics are estimated with Agisoft Metashape using RGB mean images from 26 lights for each degree.



3D fruit reconstruction

Using normals and cameras parameters obtained from 20 different viewpoints, SuperNormal [1] is used to reconstruct the complete fruit.

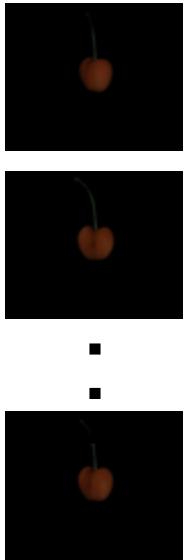


[1] Xu Cao and Takafumi Taketomi. Supernormal: Neural surface reconstruction via multi-view normal integration. pages 20581–20590, 2024.

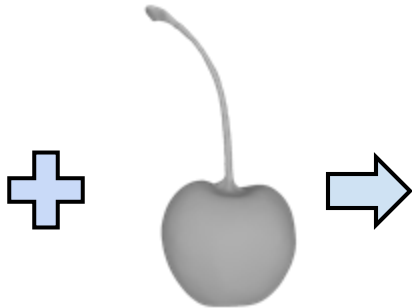
3D fruit reconstruction

Agisoft Metashape is used to texture the fruit with predicted albedos.

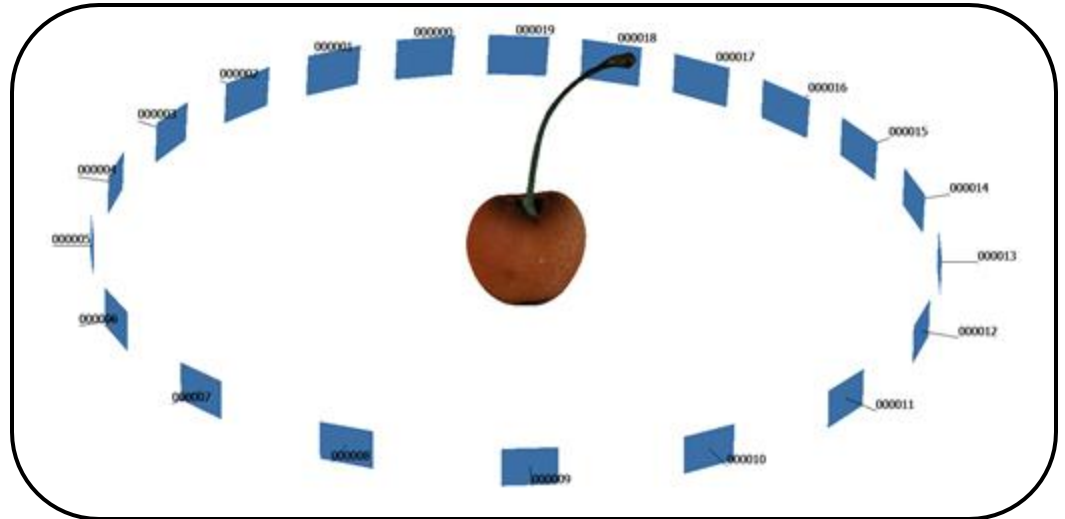
Albedos



Predicted Mesh

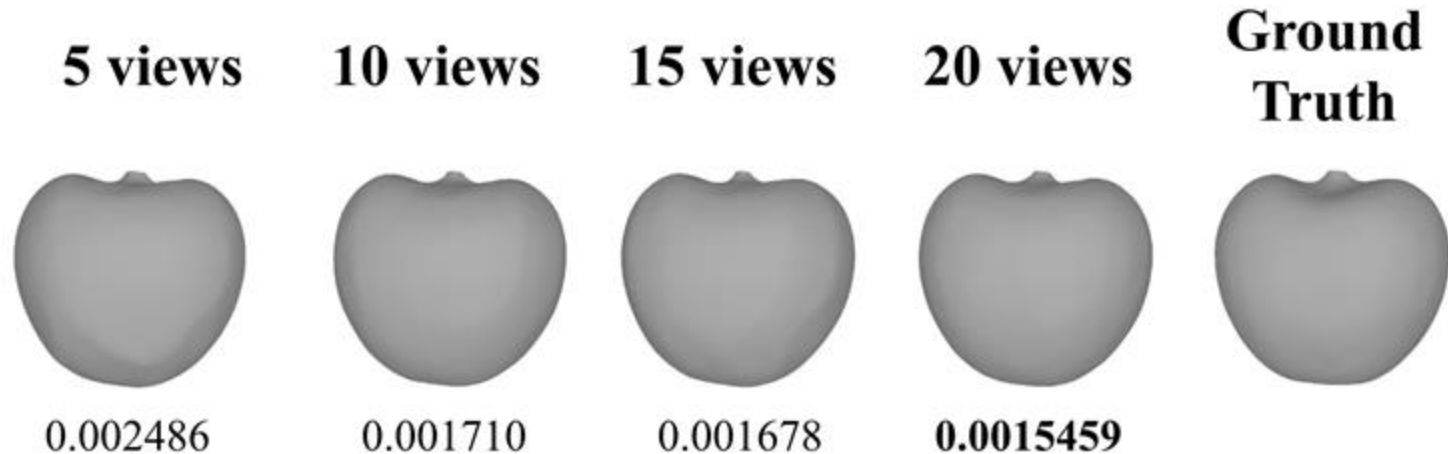


Complete Textured Fruit



3D fruit reconstruction

A reference geometry is obtained for one cherry using a commercial 3D scanning spray. The comparison is performed on the specular fruit under multiple viewpoints using Chamfer Distance



3D fruit reconstruction

Cherries

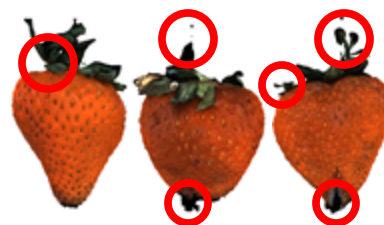
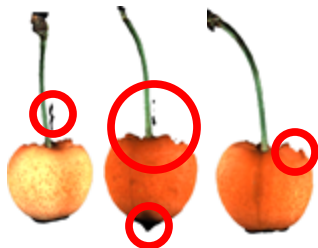
Cherry
Tomatoes

Strawberries

Nutshell

Tangerine

Metashape
(MVS)



SuperNormal
+ Agri-PS
(Ours)



Ground Truth
Albedo

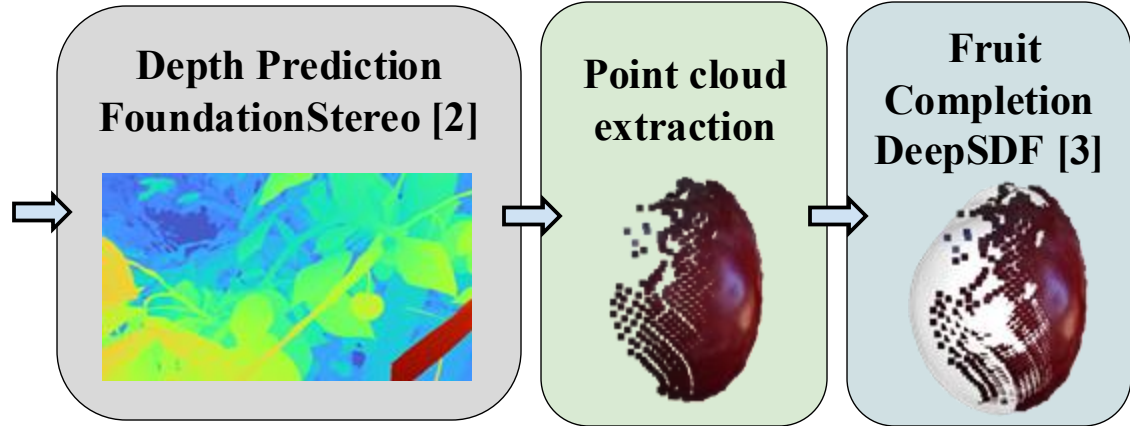


Real World Application

Left Stereo Image



Right Stereo Image



[2] Bowen Wen, Matthew Trepte, Joseph Aribido, Jan Kautz, Orazio Gallo, and Stan Birchfield. Foundationstereo: Zeroshot stereo matching. arXiv, 2025.

[3] Jeong Joon Park, Peter R. Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. Deepsdf: Learning continuous signed distance functions for shape representation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 165–174, 2019.

A Multi-View Photometric Stereo Pipeline for Specular 3D Fruit Reconstruction

Ariel Zuniga-Santana¹ - Rodrigo Verschae¹ - Gabriele Facciolo² - Shohei Nobuhara³

¹Institute of Engineering Sciences, Universidad de O'Higgins, Chile

²Universite Paris-Saclay, CNRS, ENS Paris-Saclay, Centre Borelli, France

³Institut Universitaire de France (IUF)

⁴Information and Human Sciences, Kyoto Institute of Technology, Japan